

# Dr. Sushovan Majhi

Assistant Professor of Data Science, George Washington University  
Samson Hall 313, George Washington University, Washington, D.C., USA

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## RESEARCH INTERESTS

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**Applied Topology, Topological Data Analysis (TDA), Discrete and Computational Geometry, Shape Matching, Statistical Finance, and Topological Deep Learning.**

My research primarily revolves around the interface of *mathematics*, *computer science*, and the mathematical foundations of *data science*. More specifically, I am motivated to develop provable inference techniques for data science that are inspired by topology and geometry. I also keep a keen interest in applying TDA to fascinating, real-world problems arising in fields, like finance, dynamical systems, biology, medicine, and genetics.

## EDUCATION

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- **Doctor of Philosophy in Mathematics** August 2014–December 2020  
**Tulane University**, New Orleans, LA, USA.  
*Advisor: Prof. Carola Wenk*  
*Courses:* computational geometry, computational topology, topological data analysis, differential geometry, differentiable manifolds, algorithms, data structures, computational complexity, applied mathematics, and scientific computing.
- **Master of Science in Mathematics** August 2009–May 2012  
**Tata Institute of Fundamental Research**, Bangalore, India  
*Courses:* ordinary and partial differential equations, probability theory, complex analysis, functional analysis, numerical linear algebra, measure theory, mechanics.
- **Bachelor of Science in Mathematics (Hons.)** July 2006–May 2009  
**Ramakrishna Mission Vidyamandira, Calcutta University**, West Bengal, India  
*Courses:* calculus, real analysis, linear algebra, numerical analysis, game theory, statistics, physics.

## WORK EXPERIENCE

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- **Assistant Professor** August 2024–current  
**Data Science Program, George Washington University**, Washington D.C., USA
- **Visiting Assistant Professor** August 2023–July 2024  
**Data Science Program, George Washington University**, Washington D.C., USA
- **Postdoctoral Research Fellow** January 2021–July 2023  
**School of Information, University of California**, Berkeley, USA  
*Role:* The responsibilities included conducting research broadly in data science, forging new research collaborations, organizing research webinar series.

## PREPRINTS

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8. **SM**. A constant-factor approximation of the Gromov–Hausdorff distance in the plane, June 2026. Available at: [arXiv:2606.17051](#) [cs.CG]
7. **SM**, Atish Mitra, Žiga Virk, Pramita Bagchi. A Closed-Form Adaptive-Landmark Kernel for Certified Point-Cloud and Graph Classification, May 2026. Available at: [arXiv:2605.04046](#) [cs.LG]
  - Submitted to *Journal of Machine Learning Research*
6. **SM**, Atish Mitra, Žiga Virk, Pramita Bagchi. A Closed-Form Persistence-Landmark Pipeline for Certified Point-Cloud and Graph Classification, May 2026. Available at: [arXiv:2605.02836](#) [cs.LG]
  - Submitted to *Transactions on Machine Learning Research*

5. Salam Rabindrajit Luwang, **SM**, Vishal Mandal, Atish J. Mitra, Md. Nurujjaman, Buddha Nath Sharma. Interpretable Classification of Time Series Using Euler Characteristic Surfaces, 25 pages, March 2026. Available at: [arXiv:2603.15079](https://arxiv.org/abs/2603.15079) [cs.LG]
4. Kazuhiro Kawamura, **SM**, Atish Mitra. The Shadow of Vietoris–Rips Complexes in Limits, 29 pages, January 2026. Available at: [arXiv:2601.01359](https://arxiv.org/abs/2601.01359) [math.AT]
  - Submitted to *Foundations of Computational Mathematics*
3. Joshua Dorrington, **SM**, Atish Mitra, James Moukheiber\*, Demi Qin, Jacob Sriraman\*, Kristian Strommen. Topology of the Polar Vortex and Montana Weather, 6 pages, Mar 2025. Available at: [arXiv:2503.20743](https://arxiv.org/abs/2503.20743) [math.DS]
  - Presented at ATMCS’25 & submitted to *La Matematica*
2. Halley Fritze, **SM**, Marissa Masden, Atish Mitra, Michael Stickney. Faithful Reeb Graph Reconstruction of a Tectonic Subduction Zone from Earthquake Hypocenters. Available at: [arXiv:2410.19410](https://arxiv.org/abs/2410.19410) [cs.CG]
  - Presented at ATMCS’25 & submitted to *La Matematica*
1. Rafal Komendarczyk, **SM**, Will Tran. Topological Stability and Latschev-type Reconstruction Theorems for Spaces of Curvature Bounded Above, 27 pages, June 2024. Available at: [arXiv:2406.04259](https://arxiv.org/abs/2406.04259) [math.AT]
  - Submitted to *Discrete & Computational Geometry*

## REFEREED JOURNAL PUBLICATIONS

11. Rafal Komendarczyk, **SM**, Atish Mitra. Vietoris–Rips Shadow for Euclidean Graph Reconstruction. *Journal of Applied and Computational Topology*, June 2026. Available at: [arXiv:2506.01603](https://arxiv.org/abs/2506.01603) [math.AT]
10. Buddha Nath Sharma, Anish Rai, SR Luwang, Md. Nurujjaman, **SM**. Causality Analysis of COVID-19 Induced Crashes in Stock and Commodity Markets: A Topological Perspective. *International Journal of Modern Physics C*, September 2025. DOI: [10.1142/S0129183126500221](https://doi.org/10.1142/S0129183126500221). Also available at: [arXiv:2502.14431](https://arxiv.org/abs/2502.14431) [q-fin.ST]
9. Henry Adams, Florian Frick, **SM**, Nichola McBride\*. Hausdorff VS Gromov–Hausdorff Distances. *Discrete and Computational Geometry*, February 2025. DOI: [10.1007/s00454-025-00722-9](https://doi.org/10.1007/s00454-025-00722-9) Also available at: [arXiv:2309.16648](https://arxiv.org/abs/2309.16648) [math.MG]
8. Kundan Mukhia, Anish Rai, SR Luwang, Md Nurujjaman, **SM**, Chittaranjan Hens. Complex Network Analysis of Cryptocurrency Market During Crashes. *Physica A: Statistical Mechanics and its Applications*, November 2024. DOI: [10.1016/j.physa.2024.130095](https://doi.org/10.1016/j.physa.2024.130095). Also available at: [arXiv:2405.05642](https://arxiv.org/abs/2405.05642) [q-fin.ST]
7. Anish Rai, Buddha Nath Sharma, Salam Rabindrajit Luwang, Md.Nurujjaman, **SM**. Identifying Extreme Events in the Stock Market: A Topological Data Analysis. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, October 2024. DOI: [10.1063/5.0220424](https://doi.org/10.1063/5.0220424). Also available at: [arXiv:2405.16052](https://arxiv.org/abs/2405.16052) [q-fin.ST]
6. **SM**. Demystifying Latschev’s theorem: Manifold Reconstruction from noisy data. *Discrete & Computational Geometry*, May 2024. DOI: [10.1007/s00454-024-00655-9](https://doi.org/10.1007/s00454-024-00655-9). Also available at: [arXiv:2305.17288](https://arxiv.org/abs/2305.17288) [math.AT]
5. **SM** and Carola Wenk. Distance Measures for geometric graphs. *Computational Geometry: Theory and Applications*, March 2024. DOI: [10.1016/j.comgeo.2023.102056](https://doi.org/10.1016/j.comgeo.2023.102056). Also available at: [arXiv:2209.12869](https://arxiv.org/abs/2209.12869) [cs.CG]
4. **SM**. Vietoris–Rips complexes of metric spaces near a metric graph. *Journal of Applied and Computational Topology*, May 2023. DOI: [10.1007/s41468-023-00122-z](https://doi.org/10.1007/s41468-023-00122-z). Available at: [arXiv:2204.14234](https://arxiv.org/abs/2204.14234) [math.AT]

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\* Undergraduate student at the time of research or submission

3. **SM**, Jeffrey Vitter, and Carola Wenk. Approximating Gromov-Hausdorff distance in Euclidean space. *Computational Geometry: Theory and Applications*, January 2024. DOI: [10.1016/j.comgeo.2023.102034](https://doi.org/10.1016/j.comgeo.2023.102034). Also available at: [arXiv:1912.13008](https://arxiv.org/abs/1912.13008) [math.MG]
2. Anish Rai, Ajit Mahata, Md Nurujjaman, **SM**, and Kanish Debnath. A sentiment-based modeling and analysis of stock price during the COVID-19: U- and Swoosh-shaped recovery. *Physica A: Statistical Mechanics and its Applications*. April 2022. DOI: [10.1016/j.physa.2021.126810](https://doi.org/10.1016/j.physa.2021.126810). Also available at: [arXiv:2110.03986](https://arxiv.org/abs/2110.03986) [q-fin.ST]
1. Brittany Terese Fasy, Rafal Komendarczyk, **SM**, and Carola Wenk. On the reconstruction of geodesic subspaces of  $\mathbb{R}^N$ . *International Journal of Computational Geometry & Applications*. September 2022. DOI: [10.1142/S0218195922500066](https://doi.org/10.1142/S0218195922500066). Also available at: [arXiv:1810.10144](https://arxiv.org/abs/1810.10144) [math.AT]

## REFEREED CONFERENCE PROCEEDINGS

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4. Henry Adams, **SM**, Fedya Manin, Nicolò Zava, Žiga Virk. Lower Bounding The Gromov-Hausdorff Distance In Metric Graphs. In *Proceedings of the 42nd International Symposium on Computational Geometry (SoCG)*, New Jersey, USA, June 2–5, 2026. Available at: [arXiv:2411.09182v1](https://arxiv.org/abs/2411.09182v1) [math.MG]
  - Journal version under review at *Discrete and Computational Geometry*
3. **SM**. Demystifying Latschev’s theorem: Manifold Reconstruction from noisy data. In *Proceedings of the 40th International Symposium on Computational Geometry (SoCG)*, Athens, Greece, June 11–14, 2024. Available at: <https://doi.org/10.4230/LIPIcs.SoCG.2024.73>
2. Erin Chambers, Brittany Fasy, Benjamin Holmgren\*, **SM**, and Carola Wenk. Metric and path-connectedness properties of the Fréchet distance for paths and graphs. In *Proceedings of the 34th Canadian Conference on Computational Geometry (CCCG)*, 2023. DBLP: [conf/cccg/ChambersHFMW23](https://dblp.org/pdb/journals/cccg/2023/ChambersHF23.html). Available at: [arXiv:2308.00900](https://arxiv.org/abs/2308.00900) [cs.CG]
1. **SM**. Graph mover’s distance: An efficiently computable distance measure for geometric graphs. In *Proceedings of the 34th Canadian Conference on Computational Geometry (CCCG)*, 2023. DBLP: [conf/cccg/Majhi23](https://dblp.org/pdb/journals/cccg/2023/SM23.html). Available at: [arXiv:2306.02133](https://arxiv.org/abs/2306.02133) [cs.CG]

## WORKSHOP CONTRIBUTIONS

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7. Pramita Bagchi, **SM**, Atish Mitra, Žiga Virk. Statistical Analysis of Persistence Landmarks: A Hilbert Space Embedding of the Space of Persistence Diagrams, At *Fall Workshop on Computational Geometry (FWCG)*, Queens College, 2025.
6. Enrique Alvarado, Daniela Beckelhymer, Joshua Dorrington, Tung Lam, **SM**, Jasmine Noory, María Sánchez Muniz, Kristian Strommen. Detecting the Indian Monsoon using Topological Data Analysis. At *ATMCS’25*. June 2025. Available at: [arXiv:2504.01022](https://arxiv.org/abs/2504.01022) [physics.ao-ph]
5. Halley Fritze, **SM**, Marissa Masden, Atish Mitra. Hausdorff-Approximation of Metric Graphs, At *Fall Workshop on Computational Geometry (FWCG)*, Tufts University, 2024. Available at: [arXiv:2410.19410](https://arxiv.org/abs/2410.19410) [cs.CG]
4. E. Chambers, B. Fasy, B. Holmgren\*, **SM**, and C. Wenk. Path-Connectivity of Fréchet Spaces of Graphs. *Computational Geometry: Young Researchers Forum*, 2022
3. **SM** and Carola Wenk. Distance Measures for Geometric Graphs, At *Fall Workshop on Computational Geometry (FWCG)*, 2022
2. Brittany Terese Fasy, **SM** and Carola Wenk. Threshold-based graph reconstruction using discrete Morse theory. In *Fall Workshop on Computational Geometry (FWCG)*, 2018. Abstract available at: [Link](#)
1. Brittany Terese Fasy, Rafal Komendarczyk, **SM**, and Carola Wenk. Topological reconstruction of metric graphs in  $\mathbb{R}^N$ . At *Fall Workshop on Computational Geometry (FWCG)*, 2017. Abstract available at: [Link](#)

## GRADUATE TEACHING EXPERIENCE

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- **Linear Algebra for Data Science** Summer 2024  
Data Science Program, George Washington University, USA  
*Topics:* linear algebra foundations, Gaussian elimination, pivoting techniques for stability, LU decomposition, eigenvalues and eigenvectors, solving linear systems using iterative methods, projections and linear regression, SVD and applications. *Website:* [linalg.mathematics.land](http://linalg.mathematics.land)
- **Algorithm Design (DATS 6001)** Spring 2024–current  
Data Science Program, George Washington University, USA  
*Topics:* basics of computational complexity, data structures (array, stack, queue, tree, etc), algorithms (search, sort, etc), and programming paradigms like dynamic programming and greedy algorithms.  
*Website:* [6001-majhi.datasci.land](http://6001-majhi.datasci.land)
- **Topological Data Analysis (CS)** Fall 2023  
National Institute of Technology, Sikkim, India  
*Topics:* topological spaces, metric spaces and their examples, simplicial complexes, homology in  $\mathbb{Z}/2$  coeff, persistent homology, bottleneck and Wasserstein distance, and non-linear time-series analysis using TDA.  
*Website:* [tda.mathematics.land](http://tda.mathematics.land)
- **Computer Science Foundations (DATS 6450)** Fall 2023  
Data Science Program, George Washington University, USA  
*Topics:* computer design, programming in Python, object-oriented programming
- **Introduction to Data Mining (DATS 6103)** Fall 2023–current  
Data Science Program, George Washington University, USA  
*Topics:* data wrangling, linear and logistic regression, classification, clustering, data visualization, support vector machines, machine learning algorithms.
- **Statistics for Data Science (MIDS 203)** August 2020–July 2023  
School of Information, University of California, Berkeley, USA  
*Topics:* probability theory, sampling distributions, estimators and convergence theorems, confidence intervals, hypothesis testing, and regression.
- **Linear Algebra, Complex Analysis** January 2013–June 2013  
Christ University and Scimetric Pvt Ltd, Bangalore, India
- **Analysis, Linear Algebra, Complex Analysis** November 2011–July 2012  
GATE-IIT Coaching Institute, JP Nagar, Bangalore, India  
Graduate level, for competitive national exams, e.g., National Eligibility Test
- **Analysis and Linear Algebra** February 2012–July 2012  
MES College, Department of Mathematics, Malleswaram, Bangalore, India

## UNDERGRADUATE TEACHING EXPERIENCE

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- **Data Science for All** Summer 2025  
Data Science Program, George Washington University, USA  
*Topics:* Foundations of data science for undergraduate students
- **Data Science Capstone (DATS 4001)** Fall 2024–current  
Data Science Program, George Washington University, USA  
*Goal:* The goal of the capstone is for the students to apply the knowledge acquired throughout the GWU Data Science undergraduate program to solve a real-world problem, using real data. It typically starts with developing S.M.A.R.T. questions and goes through the entire data science life cycle in multiple iterations.

- **Undergraduate Statistics for Business Students (MATH 1140)** Summer 2019  
Tulane University, USA  
*Topics:* sampling methods, descriptive statistics, probability theory, random variables, limit theorems, confidence intervals, hypothesis testing, and linear regression.

## OTHER TEACHING

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- **Graduate Teaching Assistant** Fall 2014–Spring 2017  
Tulane University, USA

## INVITED TALKS AND ACCEPTED ABSTRACTS

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- **Title:** Lower Bounding the Gromov–Hausdorff Distance in Metric Graphs June 5, 2026  
42nd International Symposium on Computational Geometry (SoCG), Rutgers University, New Brunswick, NJ
- **Title:** Geometric Latschev’s Theorem March 11, 2026  
Spring Topology Conference, University of Alabama
- **Title:** Vietoris–Rips Shadow for Euclidean Graph Reconstruction January 7, 2026  
Joint Mathematics Meetings, Washington, D.C.
- **Title:** Vietoris–Rips Shadow for Euclidean Graph Reconstruction Nov 8, 2025  
Fall Workshop on Computational Geometry, Queens College
- **Title:** Vietoris–Rips Shadow for Euclidean Graph Reconstruction Oct 4, 2025  
AMS Southeastern Sectional Meeting, Tulane University
- **Title:** Topological and Geometric Reconstruction of Spaces Beyond Manifolds June 19, 2025  
TDA Week, Kyoto University, Japan, 2025
- **Title:** Topological Stability and Latschev-type Reconstruction Theorems for  $CAT(\kappa)$  Spaces March 8, 2025  
AMS Southeastern Sectional Meeting, Clemson University
- **Title:** Topological Stability and Latschev-type Reconstruction Theorems for Spaces of Curvature Bounded Above March 6, 2025  
Spring Topology and Dynamics, Christopher Newport University, Newport News
- **Title:** Predicting the Onset and Withdrawal of the Indian Monsoon using Persistent Homology January 9, 2025  
AMS Special Session: Climate Science at the Interface Between Topological Data Analysis and Dynamical Systems Theory, Joint Mathematics Meetings, Seattle, 2025
- **Title:** Lower Bounding the Gromov–Hausdorff distance in Metric Graphs Nov 14–15, 2024  
The 31st Fall Workshop on Computational Geometry, Tufts University
- **Title:** A Taste of Topological Data Analysis (TDA): Reconstruction of Shapes Oct 14, 2024  
Indian Institute of Technology, Mandi, India
- **Title:** A Taste of Topological Data Analysis (TDA): Reconstruction of Shapes Oct 8, 2024  
Vellore Institute of Technology, Chennai, India
- **Title:** Demystifying Latschev’s Theorem June 14, 2024  
Symposium on Computational Geometry, Athens, Greece
- **Title:** Demystifying Latschev’s Theorem March 28, 2024  
Montana State University

- **Title:** Demystifying Latschev's Theorem August 23, 2023  
Applied Algebraic Topology Research Network (AATRN)  
Available on: [YouTube](#)
- **Title:** Graph Mover's Distance August 2–4, 2023  
The 34th Canadian Conference on Computational Geometry, Montreal, Canada
- **Title:** Similarity Measures for Geometric Graphs October 14–15, 2022  
The 30th Fall Workshop on Computational Geometry  
North Carolina State University, USA
- **Title:** Topological Methods in the Reconstruction and Comparison of Shapes February, 2022  
Mathematics Department, ICEAI University, Tripura, India
- **Title:** A Taste of Topological Data Analysis (TDA): Reconstruction of Shapes September, 2021  
Department of Mathematics, Hunter College, NY, USA
- **Fall Workshop on Computational Geometry** October, 2018  
Queens College, New York, USA
- **Fall Workshop on Computational Geometry** November, 2017  
SUNY (Stony Brook), New York, USA

#### PHD DISSERTATION COMMITTEE

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- Peiqi Yang, Department of Mathematics, George Washington University Summer 2025
- Will Tran, Department of Mathematics, Tulane University, Spring 2024  
*Thesis Title:* Distortion and Curvature in the Shape Reconstruction Problem

#### PHD ADVISEES

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1. Buddha Nath Sharma, PhD student at August 2025–current  
National Institute of Technology, Sikkim, India  
*Thesis Title:* Applications of topological data analysis in financial markets  
*Co-advised with* [Md. Nurujjaman](#)

#### GRADUATE STUDENT MENTEES

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- Mauricio Montes, PhD student March 2025–August 2025  
Auburn University  
*Project Title:* Statistical Analysis of Euler Characteristic Surfaces
- Khush Shah and Shikha Kumari, were data science master's students August 2023–December 2024  
George Washington University,  
*Project Title:* Matching geometric graphs using Graph Neural Networks  
*Now:* Junior Data Scientist at Cintra and AI/ML & Software Engineer at Nexas 8 International, respectively
- Anish Rai, was a PhD student at August 2022–August 2024  
National Institute of Technology, Sikkim, India  
*Project Title:* Prediction of stock market crashes using topological data analysis  
*Now:* Postdoctoral scholar at Chennai Mathematical Institute (CMI), Chennai

## UNDERGRADUATE STUDENT MENTEES

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- Abby Stein  
Senior at George Washington University August 2025–current
- Jacob Sriraman  
was at Montana State University August 2025–current  
Now: Master's student at Montana State University
- James Moukheiber  
was at George Washington University August 2024–current  
Now: Master's student at the University of Zurich

## DEPARTMENT/UNIVERSITY SERVICE

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- Chair, Undergraduate Curriculum Committee  
George Washington University, Data Science Program August 2025–current
- Undergraduate Advisor  
George Washington University, Data Science Program August 2024–current

## AWARDS, SCHOLARSHIPS, RECOGNITION

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- **Travel Grant**, University of California, Berkeley
- **UGC-CSIR NET Research Fellowship**, India, June 2012
- **TIFR Junior Research Fellowship** for pursuing Integrated PhD studies at TIFR-CAM, Bangalore, India. August 2009.
- **Secured grade “A” in SCIENCE TALENT SEARCH EXAMINATION**  
conducted by JATIYA VIJNAN PARISAD and INDIAN SCIENCE CONGRESS ASSOCIATION

## ACADEMIC SERVICES

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- I organized two special sessions on “Topological Data Analysis for Non-linear Dynamics” and “Topological and Geometric Shape Reconstruction” at the Joint Mathematics Meetings (JMM) in Washington DC, 2025.
- I co-organized a special session on “Advances in Applied Topology and Topological Data Analysis” at the AMS southeastern sectional meeting at Tulane University in 2025.
- I have been a reviewer for journals, like Discrete and Computational Geometry, Foundations of Data Science, Journal of Combinatorial Optimization, and International Journal of Computational Geometry.
- I have been a reviewer for conferences, like the International Symposium on Computational Geometry, International Symposium on Spatial and Temporal Databases, European Workshop on Computational Geometry, ACM International Conference on Advances in Geographic Information Systems, WADS Algorithms and Data Structures Symposium.
- I organized SIAM Graduate Student Chapters at Tulane University.
- I have been organizing a data science webinar series at the University of California, Berkeley.

## COMPUTATIONAL SKILLS

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Java, C, R, Python, Ruby, JavaScript, SQL, Bash.

## SOFTWARE PROJECTS

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- **Simplicial Complexes in JS** [GitHub](#)  
JavaScript implementation of some of the widely used computations on simplicial complexes. The library also implements the Smith Normal Form in order to compute the homology groups of an abstract complex.
- **Shape Reconstruction Visualization** [WebApp](#) | [GitHub](#)  
To complement my PhD research, I implemented my topological reconstruction algorithm for planar metric graphs in this library. The library is written in JavaScript and made available to users as a web-app.  
*Skills:* JavaScript, HTML, CSS.

## POPULAR MEDIA

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- Interviewed by GWU Hatchet for an [Article](#) August 2024

## ENTREPRENEURIAL EXPERIENCE

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- **Scimetric Edulabs Private Limited** December 2012–April 2017  
Bangalore, India  
*Role:* **co-founder** and **director**  
In this start-up venture, our objective was to motivate and train students in higher education. We won the franchise to work with several private colleges in India. We coached science students for standardized entrance tests for PhD and academic jobs. The company employed 6 trainers.

## HOBBY

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In my spare time, I write tutorials on *random* topics in order to make mathematics and statistics a little more interactive; they can be found here: <https://www.smajhi.com/tutorials>. I also enjoy playing the piano and classical guitar.